# Gangadhar DN

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## Professional Summary


Experienced Data Scientist with 4+ years of expertise in leveraging Generative AI (Artificial Intelligence), Large Language Models, and advanced data modeling for predictive analytics. Proven record of accomplishment in business intelligence and data analysis, particularly within the Banking and Healthcare sectors. Skilled in navigating Google Cloud and Microsoft Azure environments, seeking a challenging role to apply my skills in solving complex problems and driving data-driven decisions, contributing to innovative solutions and business success.


## Experience

### Data Scientist – Relanto — May 2024 – Present

1. **Smart Capture**
   - Developed a GCP-based, GenAI-powered platform for automated document processing & budget forecasting, enabling real-time data integration for enterprise clients.
   - Engineered a scalable document processing pipeline using Flask & Google DocAI, applying prompt tuning for 95% accurate, multi-format data extraction, reducing manual effort by 60%.
   - Integrated OAuth 2.0 & SAML for robust, enterprise-grade authentication, ensuring secure & streamlined Google and Anaplan login flows.
2. **Chatbot Sales Copilot and Learn Bot**
   - Developed AI-driven chatbot assistants (Edu/Sales Copilot) for intelligent data processing, integrating structured/unstructured data from PDFs and Amazon Redshift.
   - Optimized LLM performance & accuracy with OpenAI, Azure, Gemini models, utilizing LlamaIndex (PGVector, Chroma, Pinecone) for RAG, Redis Cache, and prompt tuning for query precision/routing.
   - Ensured robust data security via PII protection, data masking, & AWS Secret Manager/Parameter Store, enabling contextual LLM queries with SQLNode Mapping.
   - Built a scalable FastAPI backend packaged as a Python library, and implemented Langfuse for comprehensive logging/observability, ensuring reliability.

### Data Scientist – Quadratyx — Jul 2021 – May 2024

1. **Generative AI – LLM**
   - Developed Retrieval-Augmented Generation RAGs for multiple conversational tools, enhancing domain-specific data - health and financial data interactions.
   - Utilized Azure and GCP (Google Cloud) services along with cutting-edge Large Language Models including Llama2, gpt-4, Falcon-7B, Whisper & other open-source models.
   - Performed fine-tuning and evaluation of Large Language Models using LlamaIndex and Langchain frameworks, utilized PGVector from PostgreSQL for vector database.
   - Building interactive chatbots, Django, Gradio, applications and worked on several use-cases including Text-to-SQL, Utilizing PandasAI with prompts to generate visualization plots.
   - Worked on prompt engineering to improve the effectiveness of prompts.
2. **Default Payments Forecast**
   - Built multiple tree-based models with diverse approaches for claim analysis using Databricks and Azure cloud services.
   - Utilized PySpark for fast computing to handle large datasets from Azure DataLake, and efficiently through its distribution system.
   - Building end-to-end pipelines from data loading to model deployment and involved in customer discussions.

3. **Next-Gen Fraud Detection**
   - Proficient in creating PySpark scripts for performing data wrangling/munging on large transactional on-premises data using HDFS.
   - Implemented a fraud detection model achieving a 150% increase in fraud identification accuracy compared to previous systems, for retail transactional data, demonstrating a significant enhancement in risk mitigation capabilities.
   - Experienced in building automation solutions using Apache Airflow, creating DAGs for scheduling and monitoring, enhancing overall data workflow automation.
   - Developed multiple Flask API scripts to perform tasks efficiently.
4. **Prescription Automation System**
   - Developed and implemented an email automation tool to streamline daily quality analysis and classification report generation processes, enhancing efficiency and accuracy.
   - Conducted thorough manual and automated pattern analysis on medical image data, leveraging advanced techniques to extract valuable insights & improve diagnostic accuracy.
   - Executed precise data extraction and image processing tasks on image data using OCR technology, facilitating seamless integration of visual information into analytical workflows.

## Machine Learning Intern - Aryabhata Technologies Oct 20 – Mar 21

- Implemented Convolutional Neural Networks (CNN) to optimize data extraction from images, enabled data extraction from images with 85% accuracy reducing time taken by 50%
- Utilized Generative Adversarial Networks (GANs) to create synthetic data effectively, 30% more effectively than traditional data generation methods.
- Developed Neural Networks, leveraging OpenCV techniques for plant detection and segmentation. Automated Data Extraction using Scrappy.
- Utilized Pytorch, TensorFlow, Keras framework.

## Education

**Presidency University,** Bengaluru

**B.Tech, Computer Science Engineering (CSE)** **2017 - 2021**

## Skills

- **Programming Languages**: Python, C, JavaScript.
- **Frameworks/Libraries:** PySpark, LlamaIndex, Langchain, TensorFlow, PyTorch, PandasAI, Flask, Djang, Langfuse.
- **Cloud:** Microsoft Azure, Google Cloud Platform, Amazon Web Services.
- **Databases:** MongoDB, MySQL, MS SQL, PostgreSQL, Redshift.
- **Technologies/Tools**: Git, OpenCV, Power BI, Airflow, Postman, Docker, Hadoop, Streamlit, Gradio, FastAPI.

## Competitions, Certificates and Activities

- **Microsoft Azure Data Fundamentals (DP-900).**
- **Microsoft Azure AI Fundamentals (AI-900).**
- **Microsoft Azure Data Scientist Certification (DP-100).**
- **ActiveAI (Intel)–** Langchain & Vector Databases in Production.
- **GenAI 360 -** Retrieval Augmented Generation for Production with Langchain, LlamaIndex.
- **Apache Airflow** official certification.
- **Databricks** – **Generative AI**, and **Data Lakehouse Fundamentals**.
- **Kaggle** Competition – "**Zero to GANs Human Protein Classification**" – Top 10%.
- Attending major **Tech events** for learning & networking, in **Bengaluru** & **Hyderabad**.